

## Cybersecurity-Based GSM Data Protection System - Project Document (White Background Diagrams)

This document is the fully aligned project report for the **Design and Implementation of a Cybersecurity-Based GSM Data Protection System**. It includes the complete chapters, all necessary diagrams rendered as Mermaid blocks with white backgrounds, and specific screenshot placeholders corresponding to the app's current UI.

TITLE PAGE

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**DESIGN AND IMPLEMENTATION OF A CYBERSECURITY-BASED GSM DATA PROTECTION SYSTEM**

**BY [Your Full Name]**

**MATRICULATION NUMBER: [INSERT MATRIC NUMBER]**

**A PROJECT SUBMITTED TO THE DEPARTMENT OF COMPUTER SCIENCE, CRESTOAK COLLEGE OF NURSING,  
MANAGEMENT, AND TECHNOLOGY**

**IN PARTIAL FULFILMENT OF THE REQUIREMENTS FOR THE AWARD OF THE DEGREE OF BACHELOR OF SCIENCE  
(B.Sc.) IN COMPUTER SCIENCE**

**JULY, 2026**

## CERTIFICATION

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This is to certify that this project titled "Design and Implementation of a Cybersecurity-Based GSM Data Protection System" was carried out by **[Your Full Name]** with Matriculation Number **[Insert Matric Number]** in the Department of Computer Science, Crestoak College of Nursing, Management, and Technology, under my supervision. It has been examined and approved as meeting the requirements for the award of the Bachelor of Science (B.Sc.) degree in Computer Science.

**Project Supervisor**

Name: \_\_\_\_\_ Signature: \_\_\_\_\_ Date: \_\_\_\_\_

#### APPROVAL PAGE

This project titled "Design and Implementation of a Cybersecurity-Based GSM Data Protection System" has been read, examined, and approved by the Department of Computer Science, Crestoak College of Nursing, Management, and Technology, as meeting the requirements for the award of the Bachelor of Science (B.Sc.) degree in Computer Science.

Table 1: Approval Schedule

| Name               | Signature | Date  |
|--------------------|-----------|-------|
| Project Supervisor | _____     | _____ |
| Head of Department | _____     | _____ |
| External Examiner  | _____     | _____ |

## DEDICATION

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This project is dedicated to Almighty God for His grace, wisdom, strength, and guidance throughout my academic journey. It is also dedicated to my parents, family members, lecturers, and friends for their prayers, encouragement, sacrifices, and unwavering support during the successful completion of this project.

## ACKNOWLEDGEMENTS

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I sincerely thank Almighty God for His mercy, protection, wisdom, and strength throughout my undergraduate studies. His guidance made the successful completion of this project possible.

I express my heartfelt appreciation to my project supervisor, **Mr. Femi Adebayo Joseph**, for the valuable guidance, constructive suggestions, patience, and encouragement provided throughout this research.

I also appreciate the Head of the Department of Computer Science, all lecturers in the department, and every member of staff whose knowledge and support contributed to my academic development.

My sincere gratitude goes to my parents and family members for their love, prayers, financial support, and continuous encouragement throughout my studies.

I equally thank my classmates and friends for their cooperation, motivation, and support during the course of this project.

Finally, I appreciate everyone who contributed directly or indirectly to the successful completion of this research.

## ABSTRACT

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The rapid growth of Global System for Mobile Communication (GSM) technology has significantly improved communication, online banking, electronic commerce, healthcare, education, and many other digital services. However, this growth has also increased cybersecurity threats such as unauthorized access, SIM swap fraud, phishing attacks, identity theft, and data leakage. These threats have created the need for stronger methods of protecting GSM data.

This study focused on the design and implementation of a cybersecurity-based GSM data protection system. The aim of the study was to develop a secure web-based application that demonstrates how modern cybersecurity techniques can be used to improve the protection of GSM data. The system was developed using the System Development Life Cycle (SDLC) methodology.

The application was implemented using HTML5, CSS3, Tailwind CSS, JavaScript, PHP 8, and MySQL. It integrates several cybersecurity technologies, including user authentication, password hashing, AES-256 message encryption, One-Time Password (OTP) verification, session management, login monitoring, activity monitoring, threat detection, account locking, security logging, security alerts, and an admin dashboard.

System testing showed that all modules performed successfully. Users were able to register securely, log in using password and OTP verification, encrypt and decrypt GSM messages, while administrators successfully monitored user activities and security alerts through the admin dashboard. The system supports activity logging and history tracking to ensure transparency.

The study concludes that integrating multiple cybersecurity techniques into a single system provides stronger protection for GSM data than relying on a single security mechanism. The proposed system improves the confidentiality, integrity, and availability of GSM data and serves as a practical prototype for secure mobile communication.

**Keywords:** GSM, Cybersecurity, Data Protection, AES-256 Encryption, One-Time Password (OTP), User Authentication, Session Management, Threat Detection, Security Logging.



## CHAPTER ONE: INTRODUCTION

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### 1.1 Background of the Study

The Global System for Mobile Communication (GSM) is one of the most widely used mobile communication technologies in the world. It enables people to make voice calls, send Short Message Service (SMS), access the internet, and use many other mobile services. Since its introduction in Nigeria in 2001, GSM has transformed communication and has become an important part of everyday life. Today, individuals, businesses, schools, hospitals, banks, and government organizations depend on GSM networks for communication and digital services.

As the use of GSM continues to increase, more sensitive information is transmitted through mobile networks. Mobile phones are now used to exchange passwords, One-Time Passwords (OTPs), bank details, business information, medical records, and other personal data. Many financial transactions, online registrations, and business activities are also carried out through GSM-enabled devices. This makes the protection of GSM data very important.

Despite these benefits, GSM networks face many cybersecurity threats. Cybercriminals use different methods to attack mobile communication systems. Some of the common threats include SIM swap fraud, SIM cloning, phishing (smishing), malware attacks, identity theft, unauthorized access, call interception, and data leakage. These attacks can result in financial loss, privacy breaches, service disruption, and loss of public trust in mobile communication systems.

Cybersecurity is the practice of protecting computer systems, communication networks, and digital information from unauthorized access, attacks, damage, or theft. In GSM networks, cybersecurity involves the use of security measures such as user authentication, data encryption, access control, activity monitoring, threat detection, and security logging to protect information during transmission and storage.

According to the GSMA Mobile Telecommunications Security Landscape Report (2024), the rapid growth of mobile communication and digital services has increased the number and complexity of cyber threats targeting mobile networks. The report identifies threats such as phishing, smishing, SIM swap fraud, malware, ransomware, and attacks on mobile network infrastructure as major security challenges facing the telecommunications industry (GSMA, 2024).

Similarly, the National Institute of Standards and Technology (NIST) states that mobile devices have become essential tools for accessing sensitive personal and organizational information. As a result, strong security measures such as encryption, authentication, secure session management, continuous monitoring, and threat detection are necessary to protect users and organizations from cyberattacks (NIST, 2023).

Although GSM networks already use security features such as SIM authentication and network encryption, these measures alone are no longer enough to defend against modern cyberattacks. Many existing GSM security systems focus mainly on basic authentication and encryption but do not combine advanced security features such as AES-256 data encryption, One-Time Password (OTP) verification, activity monitoring, threat detection, security logging, login monitoring, account locking, and real-time security alerts into one integrated system. As a result, users remain exposed to different forms of cybercrime and unauthorized access to sensitive information.

Therefore, there is a need for a more comprehensive cybersecurity solution that integrates multiple security techniques within a single platform. This project focuses on the design and implementation of a cybersecurity-based GSM data protection system that combines user authentication, AES-256 data encryption, OTP verification, session management, activity monitoring, threat detection, login monitoring, account locking, security alerts, security logging, and an admin dashboard for system monitoring. The proposed system aims to improve the confidentiality, integrity, and availability of GSM data while reducing the risks associated with modern cyber threats.

### 1.2 Statement of the Problem

The increasing use of GSM technology has made mobile communication an important part of everyday life. People use their mobile phones to make calls, send messages, perform online banking, access healthcare services, and conduct business transactions. As more sensitive information is transmitted through GSM networks, protecting users' data has become a major concern.

Despite the security measures already available in GSM networks, cybercriminals continue to exploit weaknesses in mobile communication systems. Common attacks such as SIM swap fraud, phishing (smishing), SIM cloning, unauthorized access, malware, and identity theft can expose users' personal and financial information. Many existing systems depend mainly on basic authentication and network encryption, which may not be enough to prevent modern cyber threats.

In addition, some GSM security systems do not provide important security features such as strong AES-256 data encryption, One-Time Password (OTP) verification, activity monitoring, threat detection, security logging, login monitoring, account locking after repeated failed login attempts, and real-time security alerts. The absence of these features increases the risk of unauthorized access and data breaches.

Therefore, there is a need to develop a more secure GSM data protection system that combines these cybersecurity techniques into a single platform. This study seeks to design and implement a cybersecurity-based GSM data protection system that improves the security of mobile data and helps protect users against modern cyber threats.

### 1.3 Aim and Objectives of the Study

#### Aim of the Study

The aim of this study is to design and implement a cybersecurity-based GSM data protection system that improves the security of information transmitted through GSM networks using modern cybersecurity techniques.

#### Objectives of the Study

The specific objectives of this study are to:

- Design a secure user registration and authentication system for GSM users.
- Implement AES-256 encryption to protect sensitive GSM data during storage and transmission.
- Develop a One-Time Password (OTP) verification module to improve user authentication.
- Implement activity monitoring and security logging to record user and system activities.
- Develop a threat detection mechanism that monitors suspicious login attempts and generates security alerts.

Implement account locking and session management to prevent unauthorized access.  
Develop an admin dashboard for monitoring users, security activities, alerts, and system reports.  
Evaluate the effectiveness of the proposed system in improving GSM data protection.

#### 1.4 Research Questions

This study seeks to answer the following research questions:

How can a cybersecurity-based system improve GSM data protection?  
How can AES-256 encryption improve the security of GSM data?  
How can One-Time Password (OTP) verification strengthen user authentication?  
How can activity monitoring and security logging help detect suspicious activities?  
How can threat detection, login monitoring, security alerts, and account locking reduce unauthorized access?  
How can an admin dashboard improve the monitoring and management of GSM security activities?  
How effective is the proposed system in protecting GSM data against modern cyber threats?

#### 1.5 Significance of the Study

This study will be beneficial to different individuals and organizations:

**GSM Users:** The proposed system will help protect users' personal and financial information through secure authentication, AES-256 encryption, OTP verification, and account protection features.

**Telecommunication Companies:** The study will demonstrate how additional cybersecurity techniques such as threat detection, activity monitoring, and security alerts can strengthen GSM security.

**Banks and Business Organizations:** The system can help improve the protection of sensitive customer information transmitted through GSM networks, thereby reducing the risk of fraud and unauthorized access.

**Researchers and Students:** This study will serve as a useful reference for future research in cybersecurity, GSM security, and secure software development.

**Government and Regulatory Agencies:** The findings of this study may assist organizations responsible for information security and telecommunications in understanding the importance of stronger cybersecurity measures for GSM data protection.

#### 1.6 Scope of the Study

This study focuses on the design and implementation of a cybersecurity-based GSM data protection system. The system provides secure user registration and login, user authentication, One-Time Password (OTP) verification, AES-256 data encryption and decryption, activity monitoring, threat detection, login monitoring, activity history logging, account locking after repeated unsuccessful login tries, security alerts, session management, and a Security Control Terminal for system monitoring.

The study is limited to a simulated GSM security environment developed as a web application using PHP, MySQL, HTML, CSS, Tailwind CSS, and JavaScript. It does not connect directly to live GSM networks or telecommunication service providers.

#### 1.7 Limitations of the Study

This study has some limitations. The proposed system operates in a simulated environment and is not directly integrated with live GSM networks. The OTP verification process is simulated and does not use a real SMS gateway. The system is also limited by available time, financial resources, testing facilities, and hardware used during development. Despite these limitations, the system demonstrates how modern cybersecurity techniques can improve GSM data protection.

#### 1.8 Definition of Terms

**GSM (Global System for Mobile Communication):** A digital mobile communication technology used for voice, text messaging, and data services.

**Cybersecurity:** The practice of protecting computer systems, communication networks, and digital information from cyber threats and unauthorized access.

**Data Protection:** The process of securing information from unauthorized access, alteration, disclosure, or destruction.

**Encryption:** The process of converting readable information into an unreadable format to prevent unauthorized access.

**AES-256 Encryption:** An advanced encryption standard that uses a 256-bit key to securely protect digital information.

**Decryption:** The process of converting encrypted information back into its original readable form.

**User Authentication:** The process of verifying the identity of a user before granting access to a system.

**One-Time Password (OTP):** A temporary password that is valid for only one login session or transaction.

**Threat Detection:** The process of identifying suspicious activities or possible cyberattacks within a system.

**Activity Monitoring:** The continuous observation and recording of user and system activities.

**Activity History Logging:** The process of saving records of user and system activities for security checking.

**Session Management:** The process of securely controlling and maintaining a user's login session.

**Account Locking:** A security feature that temporarily blocks access after multiple failed login attempts.

**Security Alert:** A notification generated when suspicious or unauthorized activities are detected.

**Security Control Terminal:** A main control screen that lets an administrator monitor users, safety alerts, activity history, and system backups.

## CHAPTER TWO: LITERATURE REVIEW

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### 2.1 Introduction

This chapter reviews existing literature related to GSM communication, cybersecurity, and GSM data protection. It explains the major concepts, theories, and previous studies that are relevant to this research. The chapter also discusses important cybersecurity technologies such as user authentication, AES-256 encryption, One-Time Password (OTP) verification, session management, activity monitoring, threat detection, security logging, login monitoring, account locking, security alerts, and the use of an admin dashboard for security management.

The purpose of this chapter is to identify the strengths and weaknesses of existing GSM security systems, highlight the research gap, and provide the theoretical foundation for the design and implementation of the proposed cybersecurity-based GSM data protection system. Recent studies show that mobile communication systems continue to face increasing cyber threats, making stronger security measures necessary to protect users' information and privacy.

### 2.2 Conceptual Review

#### 2.2.1 Global System for Mobile Communication (GSM)

The Global System for Mobile Communication (GSM) is a digital mobile communication technology that allows users to make voice calls, send SMS, and access mobile data services. GSM was introduced to provide secure and reliable wireless communication across different countries. Today, GSM supports many services such as online banking, e-commerce, healthcare, education, and business communication.

As GSM services continue to grow, the amount of sensitive information transmitted through mobile networks has also increased. This has made GSM networks attractive targets for cybercriminals who attempt to steal personal and financial information through different types of attacks. According to the GSMA Security Landscape Report (2024), mobile networks continue to experience threats such as phishing, malware, SIM swap fraud, and attacks on network infrastructure.

#### 2.2.2 Cybersecurity

Cybersecurity is the practice of protecting computers, communication networks, software, and digital information from unauthorized access, attacks, damage, or theft. Its main goal is to ensure the confidentiality, integrity, and availability of information.

Modern cybersecurity uses several technologies to improve system security. These include user authentication, encryption, access control, session management, activity monitoring, threat detection, security logging, login monitoring, account locking, and security alerts. These technologies work together to reduce the risk of cyberattacks and protect sensitive information.

#### 2.2.3 GSM Data Protection

GSM data protection refers to the methods used to secure information transmitted through GSM networks. It involves protecting voice communication, SMS messages, user identities, authentication information, and other sensitive data from unauthorized access or modification.

Effective GSM data protection combines different security measures rather than relying on a single method. These measures include strong encryption, secure user authentication, One-Time Password (OTP) verification, activity monitoring, and continuous threat detection. Combining these security techniques provides stronger protection against modern cyber threats.

#### 2.2.4 User Authentication

User authentication is the process of verifying the identity of a user before access is granted to a system. It is the first line of defense against unauthorized access. Authentication is commonly performed using usernames, email addresses, phone numbers, passwords, biometric data, or One-Time Passwords (OTPs).

Modern cybersecurity systems often use multi-factor authentication (MFA), which combines two or more methods of verification to improve security. In this study, the proposed system uses user login credentials together with OTP verification to provide stronger protection against unauthorized access.

#### 2.2.5 AES-256 Encryption

Encryption is the process of converting readable information (plaintext) into an unreadable format (ciphertext) so that unauthorized users cannot understand it. Only users with the correct secret key can decrypt the encrypted information.

AES (Advanced Encryption Standard) is one of the most trusted encryption algorithms used today. AES-256 uses a 256-bit encryption key, making it highly secure against modern cyberattacks. It is widely used by banks, government agencies, healthcare organizations, and cloud service providers to protect sensitive information. In this study, AES-256 encryption is used to secure GSM data before it is stored or transmitted.

#### 2.2.6 One-Time Password (OTP) Verification

A One-Time Password (OTP) is a temporary password that can only be used once or within a short period. OTP adds an extra layer of security because even if a user's password is stolen, an attacker cannot access the account without the correct OTP.

Many online banking systems, e-commerce platforms, and secure applications use OTP verification to strengthen user authentication. In this study, OTP verification is used to confirm the identity of users before they gain access to protected system resources.

#### 2.2.7 Session Management

Session management is the process of creating, maintaining, and ending a user's login session securely. Once a user successfully logs in, the system creates a session that allows the user to access protected pages without logging in repeatedly.

A secure session management system protects session information from being stolen or misused. It also ends inactive sessions automatically after a specified period to reduce the risk of unauthorized access. The proposed system uses secure session management to protect authenticated users.

#### 2.2.8 Threat Detection

Threat detection is the process of identifying suspicious activities that may indicate a cyberattack. Modern systems continuously monitor user behavior and system events to detect possible security threats.

Examples of suspicious activities include repeated failed login attempts, unusual user behavior, unauthorized access attempts, and abnormal system activities. In this study, the system detects multiple failed login attempts and generates security alerts when suspicious activities are identified.

#### 2.2.9 Activity Monitoring

Activity monitoring involves continuously observing and recording the actions performed by users within a system. It helps administrators understand how the system is being used and identify unusual activities.

Examples of monitored activities include user registration, login, logout, encryption, decryption, OTP verification, and administrative actions. Activity monitoring improves system accountability and supports reviewing activity history when incidents occur.

#### 2.2.10 Security Logging

Security logging is the process of recording important security-related events that occur within a system. Log records provide useful information for monitoring system performance, detecting attacks, troubleshooting problems, and conducting forensic investigations after a security incident.

The proposed system records activities such as login attempts, OTP verification, encryption and decryption operations, security alerts, and administrative actions. These records help administrators monitor system security and respond quickly to suspicious activities by reviewing activity history.

#### 2.2.11 Login Monitoring

Login monitoring is the continuous observation of user login activities. It records successful and failed login attempts, login time, logout time, and other authentication events.

Login monitoring helps identify unusual login patterns that may indicate unauthorized access attempts. It also provides useful information for auditing and improving system security.

#### 2.2.12 Account Locking

Account locking is a security feature that temporarily blocks user access after a specified number of unsuccessful login tries. This reduces the risk of password guessing attacks and brute-force attacks.

In the proposed system, an account is temporarily locked after multiple unsuccessful login attempts. The system also records the event and generates a security alert for the administrator.

#### 2.2.13 Security Alerts

Security alerts are automatic notifications generated when suspicious activities are detected. They help administrators respond quickly to possible cyber threats before they become serious security incidents.

Examples of events that can generate alerts include repeated failed logins, account locking, unauthorized access attempts, and other suspicious activities detected by the system.

#### 2.2.14 Security Control Terminal (Admin Dashboard)

A central control screen displaying alerts, system health status, and active user controls.

In the proposed system, the admin dashboard (Security Control Terminal) enables administrators to monitor users, review security events, manage accounts, manage backups, and generate reports that support effective GSM data protection.

### 2.3 Theoretical Review

A theoretical review explains the theories that support a research study. This study is based on three important cybersecurity theories: the Confidentiality, Integrity, and Availability (CIA) Triad, the Defense-in-Depth Theory, and the Access Control Theory. These theories provide the foundation for protecting GSM data against cyber threats.

#### 2.3.1 Confidentiality, Integrity, and Availability (CIA) Triad

The CIA Triad is one of the most widely accepted models in cybersecurity. It is based on three security principles: confidentiality, integrity, and availability. According to the National Institute of Standards and Technology (NIST), these three principles form the foundation of information security and guide the design of secure information systems.

**Confidentiality** means that sensitive information should only be accessed by authorized users. In the proposed system, confidentiality is achieved through user authentication, AES-256 encryption, OTP verification, and session management.

**Integrity** means that information should remain accurate and should not be changed by unauthorized users. The proposed system protects data integrity through controlled user access, security logging, and activity monitoring.

**Availability** means that authorized users should be able to access information whenever it is needed. Features such as secure login, proper session management, and system monitoring help ensure that the system remains available to legitimate users.

The CIA Triad is highly relevant to this study because the proposed system is designed to protect the confidentiality, integrity, and availability of GSM data.

#### 2.3.2 Defense-in-Depth Theory

The Defense-in-Depth Theory states that no single security measure is enough to protect a system. Instead, multiple layers of security should be combined so that if one layer fails, other layers continue to provide protection. NIST describes Defense-in-Depth as a cybersecurity strategy that integrates people, technology, and operational controls across multiple security layers.

This theory supports the proposed system because it combines several security features instead of relying on only one. These features include user authentication, AES-256 encryption, OTP verification, login monitoring, threat detection, activity monitoring, account locking, security alerts, security logging, and session management. Together, these layers provide stronger protection against cyber threats affecting GSM data.

### 2.3.3 Access Control Theory

Access Control Theory focuses on ensuring that only authorized users can access specific system resources. It involves identifying users through authentication and assigning permissions based on their roles and responsibilities. According to NIST, access control is one of the most important aspects of computer security because it protects sensitive information from unauthorized access.

The proposed system applies this theory by requiring users to register, log in with valid credentials, complete OTP verification, and maintain secure login sessions before accessing protected resources. It also provides role-based access through an admin dashboard, where administrators can monitor users, review security logs, manage security alerts, and generate reports.

The combination of authentication, authorization, session management, and account locking helps ensure that only authorized users can use the system securely.

### 2.4 Empirical Review

Several researchers have studied GSM security and cybersecurity in recent years. Their findings show that mobile communication systems continue to face increasing cyber threats, making stronger security solutions necessary to protect users' information and privacy.

A report published by the GSMA in 2024 found that mobile communication networks are increasingly targeted by cybercriminals through attacks such as phishing, malware, SIM swap fraud, identity theft, and attacks on network infrastructure. The report recommends stronger authentication methods, encryption, continuous monitoring, and improved security management to reduce these threats.

The National Institute of Standards and Technology (NIST) also emphasizes that organizations should implement multiple security controls rather than depending on a single security measure. NIST recommends the use of encryption, access control, continuous monitoring, logging, and layered security to protect sensitive information against modern cyber threats.

Recent cybersecurity research published in 2026 further supports the Defense-in-Depth approach by showing that layered security provides better protection against evolving cyberattacks than relying on a single security mechanism. The study recommends combining multiple security controls to improve system resilience.

Although these studies provide valuable recommendations for improving cybersecurity, many existing GSM security solutions still focus on individual security controls instead of integrating several protective measures into one system. Most existing systems do not combine AES-256 encryption, OTP verification, user authentication, login monitoring, activity monitoring, threat detection, account locking, security alerts, security logging, and centralized administration within a single application.

This study builds on previous research by integrating these security technologies into one cybersecurity-based GSM data protection system. The proposed system provides multiple layers of protection to improve the confidentiality, integrity, and availability of GSM data while reducing the risk of unauthorized access and cyberattacks.

### 2.5 Summary of Literature Review

This chapter reviewed the concepts, theories, and previous studies related to GSM communication, cybersecurity, and GSM data protection. The conceptual review explained the major technologies used in the proposed system, including user authentication, AES-256 encryption, OTP verification, session management, activity monitoring, threat detection, security logging, login monitoring, account locking, security alerts, and the admin dashboard.

The theoretical review showed that the CIA Triad, Defense-in-Depth Theory, and Access Control Theory provide the foundation for protecting GSM data. The empirical review showed that recent studies continue to recommend stronger, layered cybersecurity solutions for mobile communication systems.

From the literature reviewed, it is clear that there is still a need for an integrated GSM security system that combines multiple cybersecurity techniques into one platform. This study addresses that need by designing and implementing a cybersecurity-based GSM data protection system that provides stronger protection for users and their information.



The process of the proposed system is shown below:

Figure 3.3: Proposed System Flow

```
%%{init: {"theme": "neutral", "themeVariables": { "background": "#ffffff", "primaryColor": "#ffffff", "edgeCc
flowchart TD
    Start([User Registration]) --> Login[User Login]
    Login --> VerifyPass{Password Correct?}
    VerifyPass -- No --> FailCount[Increment Failed Count]
    FailCount --> LockCheck{Failed Tries >= 3?}
    LockCheck -- Yes --> Lock[Lock Account & Send Admin Alert]
    LockCheck -- No --> Login
    VerifyPass -- Yes --> OTPGen[Generate 6-Digit OTP]
    OTPGen --> OTPVerify{OTP Correct & Valid?}
    OTPVerify -- No --> Login
    OTPVerify -- Yes --> Access[Access Dashboard]
    Access --> Encrypt[Encrypt Message - AES-256]
    Access --> Decrypt[UnEncrypt Message - AES-256]
    Access --> Logout([Log Out])
    style Start fill:#ffffff,stroke:#000000,stroke-width:1.5px,color:#000000
    style Login fill:#ffffff,stroke:#000000,stroke-width:1.5px,color:#000000
    style VerifyPass fill:#ffffff,stroke:#000000,stroke-width:1.5px,color:#000000
    style FailCount fill:#ffffff,stroke:#000000,stroke-width:1.5px,color:#000000
    style LockCheck fill:#ffffff,stroke:#000000,stroke-width:1.5px,color:#000000
    style Lock fill:#ffffff,stroke:#e11d48,stroke-width:1.5px,color:#e11d48
    style OTPGen fill:#ffffff,stroke:#000000,stroke-width:1.5px,color:#000000
    style OTPVerify fill:#ffffff,stroke:#000000,stroke-width:1.5px,color:#000000
    style Access fill:#ffffff,stroke:#16a34a,stroke-width:1.5px,color:#16a34a
    style Encrypt fill:#ffffff,stroke:#000000,stroke-width:1.5px,color:#000000
    style Decrypt fill:#ffffff,stroke:#000000,stroke-width:1.5px,color:#000000
    style Logout fill:#ffffff,stroke:#000000,stroke-width:1.5px,color:#000000
```

### 3.6 System Design

### 3.6.1 System Architecture

The application follows a standard Three-Tier software architecture to split presentation, business logic, and database storage, ensuring modularity and security:

### Figure 3.4: System Architecture

```

%%{init: {"theme": "neutral", "themeVariables": { "background": "#ffffff", "primaryColor": "#ffffff", "edgeC
graph TD
    Client["Presentation Layer (HTML5, Tailwind CSS, JavaScript)"] <--> Controller["Business Logic Layer (PHP
    Controller <--> PDO["Data Access Layer (PDO Prepared Statements)"]
    PDO <--> DB["Database Layer (MySQL)"]
    style Client fill:#ffffff,stroke:#000000,stroke-width:1.5px,color:#000000
    style Controller fill:#ffffff,stroke:#000000,stroke-width:1.5px,color:#000000
    style PDO fill:#ffffff,stroke:#000000,stroke-width:1.5px,color:#000000
    style DB fill:#ffffff,stroke:#000000,stroke-width:1.5px,color:#000000

```

### 3.6.2 Use Case Diagram

This diagram shows how the User and the Administrator interact with the system's functions:

Figure 3.5: Use Case Diagram

```

%%{init: { "theme": "neutral", "themeVariables": { "background": "#ffffff", "primaryColor": "#ffffff", "edgeC
graph LR
    User["User"]
    Admin["Admin"]

    subgraph System ["GSM Data Protection System"]
        U1(["Register Account"])
        U2(["Verify Password & OTP"])
        U3(["Lock Messages - AES"])
        U4(["Unlock Messages - AES"])
        U5(["View Own Events"])
        U6(["Log Out"])

        A1(["Authenticate Security Session"])
        A2(["Manage User Directory"])
        A3(["View System Health"])
        A4(["View Security Alerts Center"])
        A5(["Manage Database Backups"])
        A6(["Review Master Activity History"])
    end

    User --> U1
    User --> U2
    User --> U3
    User --> U4
    User --> U5
    User --> U6

    Admin --> A1
    Admin --> A2
    Admin --> A3
    Admin --> A4
    Admin --> A5
    Admin --> A6

    style User fill:#ffffff,stroke:#000000,stroke-width:1px
    style Admin fill:#ffffff,stroke:#000000,stroke-width:1px

```

### 3.6.3 Data Flow Diagram (DFD)

The flow of data from the user input through processes to storage tables:



Figure 3.6: Data Flow Diagram (DFD)

```

%%{init: { "theme": "neutral", "themeVariables": { "background": "#ffffff", "primaryColor": "#ffffff", "edgeC
graph TD

```

```
User([User Actor]) -- "1. Credentials" --> P1["Process 1.0: User Auth & OTP"]
P1 <--> D1[("Table: users & otp_codes")]
User -- "2. Message & Password" --> P2["Process 2.0: AES Message Encryption"]
P2 <--> D2[("Table: encrypted_messages")]
P1 -- "Log Event" --> P3["Process 3.0: Log Saving"]
P2 -- "Log Event" --> P3
P3 --> D3[("Table: activity_logs")]
Admin([Admin Actor]) -- "Query Logs & Backups" --> P4["Process 4.0: Admin Panel Control"]
P4 <--> D3
P4 <--> D4[("Table: security_alerts")]
P4 <--> Backups["Database Backup Storage"]
style User fill:#ffffff,stroke:#000000,stroke-width:1.5px,color:#000000
style Admin fill:#ffffff,stroke:#000000,stroke-width:1.5px,color:#000000
style P1 fill:#ffffff,stroke:#000000,stroke-width:1.5px,color:#000000
style P2 fill:#ffffff,stroke:#000000,stroke-width:1.5px,color:#000000
style P3 fill:#ffffff,stroke:#000000,stroke-width:1.5px,color:#000000
style P4 fill:#ffffff,stroke:#000000,stroke-width:1.5px,color:#000000
style D1 fill:#ffffff,stroke:#000000,stroke-width:1.5px,color:#000000
style D2 fill:#ffffff,stroke:#000000,stroke-width:1.5px,color:#000000
style D3 fill:#ffffff,stroke:#000000,stroke-width:1.5px,color:#000000
style D4 fill:#ffffff,stroke:#000000,stroke-width:1.5px,color:#000000
style Backups fill:#ffffff,stroke:#000000,stroke-width:1.5px,color:#000000
```

### 3.6.4 System Flowchart

This flowchart shows the detailed steps of user access and threat assessment:

Figure 3.7: System Flowchart

```

%{init: {"theme": "neutral", "themeVariables": { "background": "#ffffff", "primaryColor": "#ffffff", "edgeC
flowchart TD

```

```

Start([Start]) --> Login[Login[Input Email & Password]]
Login --> AuthCheck{Credentials Valid?}
AuthCheck -- No --> LockIncr[Add Failed Count]
LockIncr --> CountCheck{Count >= 3?}
CountCheck -- Yes --> LockAcc[Lock User Account & Trigger Security Alert] --> End([Stop])
CountCheck -- No --> Login
AuthCheck -- Yes --> OTPCode[Enter 6-Digit Code]
OTPCode --> OTPCheck{OTP Match & Code Valid?}
OTPCheck -- No --> AuthCheck
OTPCheck -- Yes --> InitSession[Initialize User Session]
InitSession --> Dash[Show User Dashboard]
Dash --> Action{User Action?}
Action -- Lock --> LockMsg[Encrypt using AES-256-CBC] --> Dash
Action -- Unlock --> UnlockMsg[Decrypt using AES-256-CBC] --> Dash
Action -- Logout --> DestroySession[Destroy Session] --> End

style Start fill:#ffffff,stroke:#000000,stroke-width:1.5px,color:#000000
style Login fill:#ffffff,stroke:#000000,stroke-width:1.5px,color:#000000
style AuthCheck fill:#ffffff,stroke:#000000,stroke-width:1.5px,color:#000000
style LockIncr fill:#ffffff,stroke:#000000,stroke-width:1.5px,color:#000000
style CountCheck fill:#ffffff,stroke:#000000,stroke-width:1.5px,color:#000000
style LockAcc fill:#ffffff,stroke:#e11d48,stroke-width:1.5px,color:#e11d48
style OTPCode fill:#ffffff,stroke:#000000,stroke-width:1.5px,color:#000000
style OTPCheck fill:#ffffff,stroke:#000000,stroke-width:1.5px,color:#000000
style InitSession fill:#ffffff,stroke:#000000,stroke-width:1.5px,color:#000000
style Dash fill:#ffffff,stroke:#000000,stroke-width:1.5px,color:#000000
style Action fill:#ffffff,stroke:#000000,stroke-width:1.5px,color:#000000
style LockMsg fill:#ffffff,stroke:#000000,stroke-width:1.5px,color:#000000
style UnlockMsg fill:#ffffff,stroke:#000000,stroke-width:1.5px,color:#000000
style DestroySession fill:#ffffff,stroke:#000000,stroke-width:1.5px,color:#000000
style End fill:#ffffff,stroke:#000000,stroke-width:1.5px,color:#000000

```

### 3.7 Database Design

The database structure consists of tables that store user records, message parameters, and logs. Below is the Entity Relationship Diagram (ERD):

Figure 3.8: Entity Relationship Diagram (ERD)

```
%{init: {"theme": "neutral", "themeVariables": { "background": "#ffffff", "primaryColor": "#ffffff", "edgeCcolorDiagram
```

```
USERS ||--o{ OTP_CODES : generates
USERS ||--o{ ENCRYPTED_MESSAGES : locks
USERS ||--o{ ACTIVITY_LOGS : records
USERS ||--o{ SECURITY_ALERTS : triggers
USERS ||--o{ SESSIONS : establishes
```

```
USERS {
    int user_id PK
    string full_name
    string email
    string phone_number
    string password
    string account_status
    datetime created_at
}
```

```
ADMIN {
    int admin_id PK
    string username
    string email
    string password
    datetime created_at
}
```

```
OTP_CODES {
    int otp_id PK
    int user_id FK
    string otp_code
    datetime expires_at
    string status
}
```

```

ENCRYPTED_MESSAGES {
    int message_id PK
    int user_id FK
    string original_message
    string encrypted_message
    datetime encryption_time
}

```

```
ACTIVITY_LOGS {
    int log_id PK
    int user_id FK
    string activity
    string ip_address
    datetime activity_time
}
```

```
SECURITY_ALERTS {
    int alert_id PK
    int user_id FK
    string alert_type
    string description
    datetime alert_time
    string status
}
```

```
SESSIONS {
    int session_id PK
    int user_id FK
    string session_token
}
```

```
datetime login_time  
datetime logout_time  
string session_status  
}
```

### 3.8 Input Design

The inputs of the application were designed using clean web forms styled with Tailwind CSS to ensure compatibility:

**Registration Form:** Gathers Full Name, Email, Phone Number, Password, and Password Confirmation.

**Login Form:** Gathers Email/Phone and Password.

**OTP Form:** Consists of 6 separate, single-character digit inputs that support pasting and auto-submission.

**Message Encryption Form:** Gathers recipient phone number, encryption key password, simulated network address, silent message options, and plaintext message body.

### 3.9 Output Design

The output elements provide clear details in Grade 8 English:

**User Dashboard Panel:** Shows the welcome panel, active session Threat Level Score card, list of recent events, and the lock results panel displaying Ciphertext, IV, Salt, and Signature fields.

**Security Control Terminal:** Displays administrative statistics, live System Health parameters, user status lists, backups snapshots list, and the Master Activity History table.

### 3.10 Security Design

The security architecture features multi-layered controls:

**Hashed Credentials:** Passwords are saved as bcrypt hashes using PHP's `password_hash()` function.

**Prepared SQL:** Prevents SQL Injection (SQLi) attacks by executing all database commands via PDO prepared statements.

**Cross-Site Protection:** Forms use CSRF tokens to verify submission sources and strip malicious scripts to prevent XSS.



### 3.11 System Algorithm

#### Algorithm 1: Two-Step Login with Automated Submission

User inputs login details. If credentials are correct, generate a 6-digit OTP and direct the user to the OTP page.

The UI listens to character inputs in the 6 digit boxes.

If a key is typed: focus moves to the next digit box.

If paste is detected: extract the first 6 digits, fill the boxes, and trigger submission.

If 6 digits are fully entered: combine the digits and automatically send the verification request.

#### Algorithm 2: AES-256 Message Encryption

User provides message plaintext, password, and settings.

Generate a random 16-byte initialization vector (IV) and a 16-byte cryptographic salt.

Derive a key from the password and salt. Encrypt the plaintext using AES-256-CBC.

Create an HMAC signature of the ciphertext to ensure it cannot be tampered with.

Save records and display Ciphertext, IV, Salt, and Signature to the user.

### 3.12 Development Tools and Technologies

**Frontend:** HTML5, CSS3, Tailwind CSS, JavaScript (ES6), Lucide Icons.

**Backend:** PHP 8.x MVC runtime architecture.

**Database:** MySQL 8.x database server.

**Server:** XAMPP local environment.

### 3.13 System Requirements

**Hardware:** Dual-Core CPU, 4GB RAM minimum, 10GB free disk space.

**Software:** Windows 10/11 operating system, modern web browser (Chrome, Edge, Firefox).

CHAPTER FOUR: SYSTEM IMPLEMENTATION, TESTING AND RESULTS

4.1 Introduction

This chapter presents the implementation, testing, and results of the proposed cybersecurity-based GSM data protection system. It demonstrates how the different modules work together to protect GSM data.

4.2 Development Environment

The software and hardware environments used are detailed in the tables below:

Table 4.1: Hardware Used

| Hardware Component | Specification   |
|--------------------|-----------------|
| Computer           | Laptop Computer |
| Processor          | Intel Core i5   |
| Memory (RAM)       | 8 GB            |
| Storage            | 500 GB SSD      |

Table 4.2: Software Used

| Software           | Purpose                      |
|--------------------|------------------------------|
| Windows 11         | Operating System             |
| Visual Studio Code | Source Code Editor           |
| XAMPP              | Local Web Server             |
| PHP 8              | Backend Programming Language |
| MySQL              | Database Management System   |
| Tailwind CSS       | Frontend Styling Framework   |

4.3 System Implementation

4.3.1 Home Page

The Home Page provides access links to register or log in.

+-----+  
| [Figure 4.1: Home Page Screenshot (White Background UI)] |  
| (Insert screenshot showing the clean system landing page) |  
+-----+

4.3.2 User Registration

New users register by entering their name, email, phone number, and password.

+-----+  
| [Figure 4.2: Registration Page Screenshot (White Background UI)] |  
| (Insert screenshot of the Tailwind CSS registration form) |  
+-----+

4.3.3 Secure Login

Registered users log in using their email or phone number and password.

+-----+  
| [Figure 4.3: Login Page Screenshot (White Background UI)] |  
| (Insert screenshot of the user login page) |  
+-----+

4.3.4 OTP Verification

The system displays a simulated GSM phone mockup on the right side of the screen containing the temporary 6-digit verification code. The user enters the code to log in.

+-----+  
| [Figure 4.4: OTP Verification Screen Screenshot (White Handset UI)] |  
| (Insert screenshot showing the 6 input boxes, simulated handset, |  
| and countdown timer. Hover over a box to show the tooltip) |  
+-----+

4.3.5 User Dashboard

The central dashboard showing the safety panel interface.

+-----+  
| [Figure 4.5: User Dashboard Main Panel Screenshot (White Theme UI)] |  
| (Insert screenshot showing Threat Level Score, AI Safety Tips, |  
| and the recent events history list) |  
+-----+

4.3.6 AES-256 Message Encryption

Users enter message parameters to generate the cryptographic locked ciphertext.

+-----+  
| [Figure 4.6: Message Encryption Form and Result (White Theme UI)] |  
| (Insert screenshot of Encrypt Message tab with filled-in data and green |  
| output boxes displaying Ciphertext, IV, Salt, and Signature) |  
+-----+

4.3.7 Message Decryption

Paste ciphertext and keys to unlock the hidden message content.

+-----+  
| [Figure 4.7: Message Decryption Form and Result (White Theme UI)] |  
| (Insert screenshot of UnEncrypt Message tab displaying the original |  
| unlocked plaintext message below) |  
+-----+

4.3.8 Activity Logging

Logs are saved dynamically to record user actions.

```
+-----+
| [Figure 4.8: Activity Logs History Screenshot (White Theme UI)] |
| (Insert screenshot of the Recent Events history list)           |
+-----+
```

#### 4.3.9 Threat Detection & Account Locking

Suspicious actions (like entering wrong passwords three times) lock accounts and trigger administrator warnings.

```
+-----+
| [Figure 4.9: Account Locked Warning Screenshot (White Theme UI)] |
| (Insert screenshot of the account locked warning alert message)   |
+-----+
```

#### 4.3.10 Security Control Terminal (Admin Dashboard)

The centralized system administration panel.

```
+-----+
| [Figure 4.10: Security Control Terminal Admin Page (White Theme UI)] |
| (Insert screenshot of the full admin dashboard showing stats cards, |
| System Health, backups, active users list, and activity log)         |
+-----+
```

#### 4.4 Database Implementation

The database tables are shown below:

Table 4.3: Users Table Structure

| Field          | Data Type    | Key / Constraint                          | Description                       |
|----------------|--------------|-------------------------------------------|-----------------------------------|
| user_id        | INT          | PRIMARY KEY, AUTO_INCREMENT               | Unique user identifier            |
| full_name      | VARCHAR(100) | NOT NULL                                  | User's full name                  |
| email          | VARCHAR(100) | NOT NULL, UNIQUE                          | User's email address              |
| phone_number   | VARCHAR(20)  | NOT NULL, UNIQUE                          | User's phone number               |
| password       | VARCHAR(255) | NOT NULL                                  | Hashed password                   |
| account_status | VARCHAR(20)  | CHECK (status IN ('active', 'suspended')) | Status of user account            |
| created_at     | DATETIME     | DEFAULT CURRENT_TIMESTAMP                 | Date and time account was created |

Table 4.4: Admins Table Structure

| Field      | Data Type    | Key / Constraint            | Description                       |
|------------|--------------|-----------------------------|-----------------------------------|
| admin_id   | INT          | PRIMARY KEY, AUTO_INCREMENT | Unique admin record ID            |
| username   | VARCHAR(50)  | NOT NULL, UNIQUE            | Admin username                    |
| email      | VARCHAR(100) | NOT NULL, UNIQUE            | Admin email address               |
| password   | VARCHAR(255) | NOT NULL                    | Hashed password                   |
| created_at | DATETIME     | DEFAULT CURRENT_TIMESTAMP   | Date and time account was created |

Table 4.5: OTP Codes Table Structure

| Field      | Data Type    | Key / Constraint                      | Description                                      |
|------------|--------------|---------------------------------------|--------------------------------------------------|
| otp_id     | INT          | PRIMARY KEY, AUTO_INCREMENT           | Unique OTP record identifier                     |
| user_id    | INT          | FOREIGN KEY REFERENCES USERS(user_id) | Reference to the owner user                      |
| otp_code   | VARCHAR(255) | NOT NULL                              | Hashed one-time verification code                |
| expires_at | DATETIME     | NOT NULL                              | Expiration timestamp of the OTP                  |
| status     | VARCHAR(20)  | NOT NULL                              | Status of the code (e.g., active, used, expired) |

Table 4.6: Encrypted Messages Table Structure

| Field             | Data Type | Key / Constraint                      | Description                          |
|-------------------|-----------|---------------------------------------|--------------------------------------|
| message_id        | INT       | PRIMARY KEY, AUTO_INCREMENT           | Unique message identifier            |
| user_id           | INT       | FOREIGN KEY REFERENCES USERS(user_id) | Reference to the sender user         |
| original_message  | TEXT      | NOT NULL                              | Plaintext of the original message    |
| encrypted_message | TEXT      | NOT NULL                              | AES-256 encrypted ciphertext         |
| encryption_time   | DATETIME  | DEFAULT CURRENT_TIMESTAMP             | Timestamp when encryption took place |

Table 4.7: Activity Logs Table Structure

| Field         | Data Type    | Key / Constraint                      | Description                                     |
|---------------|--------------|---------------------------------------|-------------------------------------------------|
| log_id        | INT          | PRIMARY KEY, AUTO_INCREMENT           | Unique log entry identifier                     |
| user_id       | INT          | FOREIGN KEY REFERENCES USERS(user_id) | Reference to user who performed action          |
| activity      | VARCHAR(100) | NOT NULL                              | Type of action performed (e.g., login, encrypt) |
| ip_address    | VARCHAR(45)  | NOT NULL                              | Client IP address of the requester              |
| activity_time | DATETIME     | DEFAULT CURRENT_TIMESTAMP             | Timestamp when activity occurred                |

Table 4.8: Security Alerts Table Structure

| Field       | Data Type   | Key / Constraint                      | Description                                        |
|-------------|-------------|---------------------------------------|----------------------------------------------------|
| alert_id    | INT         | PRIMARY KEY, AUTO_INCREMENT           | Unique security alert identifier                   |
| user_id     | INT         | FOREIGN KEY REFERENCES USERS(user_id) | Reference to user triggering the alert             |
| alert_type  | VARCHAR(50) | NOT NULL                              | Type of security event (e.g., failed logins, SQLi) |
| description | TEXT        | NOT NULL                              | Details and telemetry of the threat                |
| alert_time  | DATETIME    | DEFAULT CURRENT_TIMESTAMP             | Timestamp when alert was logged                    |
| status      | VARCHAR(20) | DEFAULT 'open'                        | Status of alert (e.g., open, resolved, ignored)    |

Table 4.9: Sessions Table Structure

| Field          | Data Type    | Key / Constraint                      | Description                                       |
|----------------|--------------|---------------------------------------|---------------------------------------------------|
| session_id     | INT          | PRIMARY KEY, AUTO_INCREMENT           | Unique session identifier                         |
| user_id        | INT          | FOREIGN KEY REFERENCES USERS(user_id) | Reference to the logged-in user                   |
| session_token  | VARCHAR(255) | NOT NULL, UNIQUE                      | Secure random session token                       |
| login_time     | DATETIME     | DEFAULT CURRENT_TIMESTAMP             | Timestamp when session was established            |
| logout_time    | DATETIME     | NULL                                  | Timestamp when user logged out                    |
| session_status | VARCHAR(20)  | DEFAULT 'active'                      | Current state (e.g., active, expired, terminated) |



#### 4.5 System Testing

All test cases passed during system verification.

Table 4.11: System Testing Results

| Test Case         | Expected Result               | Actual Result | Status |
|-------------------|-------------------------------|---------------|--------|
| Registration      | Account created successfully  | Successful    | Pass   |
| Login Auth        | Authenticated & redirected    | Successful    | Pass   |
| OTP Check         | Access granted to dashboard   | Successful    | Pass   |
| Encrypt Message   | Plaintext converted to AES    | Successful    | Pass   |
| UnEncrypt Message | Ciphertext converted to plain | Successful    | Pass   |
| Account Lock      | Locked after 3 failed tries   | Successful    | Pass   |
| Admin Controls    | System stats and logs visible | Successful    | Pass   |

## CHAPTER FIVE: SUMMARY, CONCLUSION AND RECOMMENDATIONS

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### 5.1 Introduction

This chapter presents the summary of the study, conclusion, and recommendations.

### 5.2 Summary of the Study

This project successfully implemented a cybersecurity-based GSM data protection system using Tailwind CSS, PHP 8, and MySQL. It integrates user authentication, password hashing, AES-256 encryption, OTP verification, session management, activity logging, threat detection, account locking, and an admin terminal (Security Control Terminal). Testing confirmed all modules run correctly.

### 5.3 Conclusion

Layered security methods provide stronger protection for GSM messages and user details than a single protection method. The prototype successfully demonstrates this concept and is ready for academic demonstration.

### 5.4 Recommendations

Communication systems should use layered cybersecurity.

OTP multi-factor authentication should be used for online services.

Strong encryption (AES-256) should be implemented to secure messages.

## APPENDICES

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### Appendix A: User Manual

Set up XAMPP locally and start Apache and MySQL.

Place the project folder in XAMPP's `htdocs/gsm-security` directory.

Open `http://localhost/gsm-security` in your web browser.

Register a new user account, enter the password, and log in.

View the simulated OTP on the handset screen and enter it.

Start locking (encrypting) and unlocking (decrypting) messages.

### Appendix B: System Screenshots

*(Take screenshots from the active app running on `http://localhost/gsm-security` and replace the placeholders in Figure 4.1 to Figure 4.10 inside this appendix. Ensure they are captured using a white browser theme/background where possible to match this document).*